ASSIGNMENT 05
OBJECT ORIENTED PROGRAMMING

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Course : Object oriented programming with C++
Group : 1MCA3

1. **Create a class Print with overloaded display() functions to display an integer, a double value, and a string.**

```
#include <iostream>
#include <string>
using namespace std;
class Print {
public:
    void display(int a)    {
        cout << "Integer: " << a << endl;
    }
    void display(double b)    {
        cout << "Double: " << b << endl;
    }
    void display(string c)    {
        cout << "String: " << c << endl;
    }
};
int main() {
    Print obj;
    obj.display(10);
    obj.display(12.5);
    obj.display(string("Hello"));
    return 0;
}
```

```
Integer: 10
Double: 12.5
String: Hello

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Process exited after 0.2518 seconds with return value 0
Press any key to continue . . .
```

2. **Create a class Area with overloaded calculate() functions to calculate the area of a square, rectangle, and circle.**

```
#include <iostream>
using namespace std;
class Area {
public:
    void calculate(int side)    {
        cout << "Area of Square = " << side * side << endl;
    }
    void calculate(int length, int breadth)    {
        cout << "Area of Rectangle = " << length * breadth << endl;
    }
    void calculate(double radius)    {
```

```
        cout << "Area of Circle = " << 3.14 * radius * radius << endl;
    }
};
int main() {
    Area obj;
    obj.calculate(4);
    obj.calculate(5, 3);
    obj.calculate(2.5);
    return 0;
}
```

```
Area of Square = 16
Area of Rectangle = 15
Area of Circle = 19.625

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```

3. **Calculate Sum**

Create a class Calculator with overloaded functions named sum():

- **sum(int a, int b) — calculate the sum of two integers.**
- **sum(int a, int b, int c) — calculate the sum of three integers.**
- **sum(double a, double b) — calculate the sum of two decimal numbers.**

Demonstrate all three overloaded functions using an object of the class.

```
#include <iostream>
using namespace std;
class Calculator {
public:
    int sum(int a, int b)    {
        return a + b;
    }
    int sum(int a, int b, int c)    {
        return a + b + c;
    }
    double sum(double a, double b)    {
        return a + b;
    }
};
int main() {
    Calculator obj;
    cout << "Sum of two integers = " << obj.sum(10, 20) << endl;
    cout << "Sum of three integers = " << obj.sum(10, 20, 30) << endl;
    cout << "Sum of decimal numbers = " << obj.sum(2.5, 3.5) << endl;
    return 0;
}
```

```
Sum of two integers = 30
Sum of three integers = 60
Sum of decimal numbers = 6

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```

4. **Create a class with overloaded result() functions to display marks in one subject, calculate the average of two subjects, and calculate the average of three subjects.**

```
#include <iostream>
using namespace std;
class Student {
public:
    void result(float a)    {
        cout << "Marks in one subject = " << a << endl;
    }
    void result(float a, float b)    {
        cout << "Average of two subjects = " << (a + b) / 2 << endl;
    }
    void result(float a, float b, float c)    {
        cout << "Average of three subjects = " << (a + b + c) / 3 << endl;
    }
};
int main() {
    Student obj;
    obj.result(85);
    obj.result(80, 90);
    obj.result(70, 80, 90);
    return 0;
}
```

```
Marks in one subject = 85
Average of two subjects = 85
Average of three subjects = 80

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```

5. **Find the Maximum (Moderate)**

Create a class Maximum with overloaded functions named findMax():

- **findMax(int a, int b) — find the greater of two integers.**
- **findMax(int a, int b, int c) — find the greatest of three integers.**
- **findMax(double a, double b) — find the greater of two decimal numbers.**

```
#include <iostream>
using namespace std;
class Maximum {
public:
    int findMax(int a, int b)    {
        if (a > b)
            return a;
        else
            return b;
    }
    int findMax(int a, int b, int c)    {
        if (a >= b && a >= c)
            return a;
        else if (b >= a && b >= c)
            return b;
```

```
    double findMax(double a, double b)    {
        if (a > b)
            return a;
        else
            return b;
    }
};
int main() {
    Maximum obj;
    cout << "Maximum of two integers = " << obj.findMax(10, 20) << endl;
    cout << "Maximum of three integers = " << obj.findMax(10, 30, 20) << endl;
    cout << "Maximum of two decimals = " << obj.findMax(5.5, 8.5) << endl;
    return 0;
}
```

```
Maximum of two integers = 20
Maximum of three integers = 30
Maximum of two decimals = 8.5

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Process exited after 0.1223 seconds with return value 0
Press any key to continue . . .
```

4. **Shopping Bill Generator**

Create a class Bill with overloaded functions named calculateBill() to calculate:

- **- The bill for one item using quantity and unit price.**
- **- The bill for two items using their quantities and prices.**
- **- The bill for a bill amount after applying a discount percentage.**

Use suitable parameter lists and return the final bill amount.

```
#include <iostream>
using namespace std;
class Bill {
public:
    double calculateBill(int quantity, double price)    {
        return quantity * price;
    }
    double calculateBill(int q1, double p1, int q2, double p2)    {
        return (q1 * p1) + (q2 * p2);
    }
    double calculateBill(double amount, double discount)    {
        return amount - (amount * discount / 100);
    }
};
int main() {
    Bill obj;
    cout << "Bill for one item = "<< obj.calculateBill(2, 100.0) << endl;
    cout << "Bill for two items = "<< obj.calculateBill(2, 100.0, 3, 50.0) << endl;
    cout << "Bill after discount = " << obj.calculateBill(500.0, 10.0) << endl;
    return 0;
}
```

```
Bill for one item = 200
Bill for two items = 350
Bill after discount = 450

--------------------------------
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Press any key to continue . . .
```